

MCQ (20 Questions)

1-The instrument used to separate components based on density is:

- A) Microscope**
- B) Pipette**
- C) Centrifuge**
- D) Balance**

2-The force used in centrifugation is:

- A) Gravity**
- B) Centrifugal force**
- C) Magnetic force**
- D) ضغط**

3-Which is more accurate?

- A) RPM**
- B) RCF**
- C) Speed**
- D) Time**

4-Microcentrifuge is used for volumes:

- A) Very large**
- B) 0.5–2 mL**
- C) >100 mL**
- D) Any volume**

5-Ultracentrifuge can reach:

- A) 10,000 RPM**
- B) 20,000 RPM**
- C) 150,000 RPM**
- D) 5,000 RPM**

6-Pipette is mainly used to:

- A) Measure weight**
- B) Transfer liquids**
- C) Heat samples**
- D) Separate components**

7-The most accurate pipette is:

- A) Pasteur**
- B) Micropipette**
- C) Serological**
- D) Graduated**

8-Air displacement pipette works by:

- A) Direct pressure**
- B) Air cushion**
- C) Heat**
- D) Magnetism**

9-Positive displacement pipette:

- A) Contains air**
- B) Piston outside liquid**
- C) Piston in direct contact**
- D) No piston**

10-A common pipetting error is:

- A) Slow operation**
- B) Air bubbles**
- C) Correct tip use**
- D) Proper storage**

11-A microscope is used to:

- A) Heat samples
- B) Magnify objects
- C) Weigh samples
- D) Separate liquids

12-Eyepiece magnification is usually:

- A) 4×
- B) 10×
- C) 40×
- D) 100×

13-Highest magnification objective is:

- A) 4×
- B) 10×
- C) 40×
- D) 100×

14-Coarse focus is used with:

- A) High power
- B) Low power
- C) Oil immersion
- D) None

15-Fine focus is used with:

- A) Low power
- B) High power
- C) Scanning
- D) None

16-Condenser function is:

- A) Magnification

B) Focus light

C) Move slide

D) Support

17-Iris diaphragm controls:

A) Magnification

B) Light amount

C) Slide movement

D) Lens rotation

18-Resolution means:

A) Size of image

B) Clarity of image

C) Color

D) Speed

19-Microscope should be carried:

A) One hand

B) By lens

C) Two hands

D) By base only

20-In centrifuge operation, you must:

A) Not balance tubes

B) Open lid

C) Balance tubes

D) Always use max speed

Answers:

1-C, 2-B, 3-B, 4-B, 5-C, 6-B, 7-B, 8-B, 9-C, 10-B

11-B, 12-B, 13-D, 14-B, 15-B, 16-B, 17-B, 18-B, 19-C, 20-C

True / False (20 Questions)

1. Centrifuge separates based on density ✓
2. RPM is more accurate than RCF ✗
3. Microcentrifuge handles small volumes ✓
4. Ultracentrifuge has low speed ✗
5. Pipettes transfer liquids ✓
6. Pasteur pipette is highly accurate ✗
7. Air displacement uses air ✓
8. Positive displacement has no air ✓
9. Air bubbles do not affect accuracy ✗
10. Higher speed is always better ✗
11. Microscope magnifies objects ✓
12. Eyepiece is usually 10× ✓
13. Coarse focus is for high magnification ✗
14. Fine focus gives precision ✓
15. Condenser focuses light ✓
16. Iris diaphragm only increases light ✗
17. Resolution is more important than magnification ✓
18. Lenses can be touched ✗
19. Microscope should be covered ✓
20. Pipettes need no maintenance ✗

Fill in the Blanks (20 Questions)

1. Centrifuge separates based on _____
2. The force used is _____
3. RCF stands for _____
4. Micropipette is used for _____ volumes

5. Pasteur pipette is _____ accurate
6. Air displacement depends on _____
7. Positive displacement has no _____
8. A common error is _____
9. Microscope is used to see _____
10. Eyepiece magnification is _____
11. Highest objective is _____
12. Coarse focus is used in _____ power
13. Fine focus is used in _____ power
14. Condenser focuses _____
15. Iris diaphragm controls _____
16. Resolution means _____
17. Microscope is carried with _____
18. Lenses cleaned by _____
19. Centrifuge requires _____
20. Pipette should not exceed _____

Answers:

1-density

2-centrifugal force

3-relative centrifugal force

4-small

5-not

6-air

7-air

8-air bubbles

9-small objects

10-10×

11-100×

12-low

13-high

14-light

15-light

16-clarity

17-two hands

18-lens paper

19-balancing

20-volume limit